```
import numpy as np
import random
import pandas as pd

jobs = {
    0: [(0, 3), (1, 2), (2, 4)],
    1: [(1, 2), (2, 3), (0, 2)],
    2: [(2, 2), (0, 4), (1, 3)]
}

num_jobs = len(jobs)
num_machines = 3

num_ants = 20
iterations = 100

alpha = 1
beta = 2
rho = 0.1
Q = 100

pheromone = np.ones((num_jobs + 1, num_jobs))

best_schedule = None
best_makespan = float('inf')

random.seed(42)
np.random.seed(42)


def generate_schedule():

    job_time = [0] * num_jobs
    machine_time = [0] * num_machines
    operation = [0] * num_jobs

    schedule = []
    sequence = []
    previous = num_jobs

    while any(operation[j] < len(jobs[j]) for j in range(num_jobs)):

        feasible = [
            j for j in range(num_jobs)
            if operation[j] < len(jobs[j])
        ]

        probability = []

        for j in feasible:

            machine, duration = jobs[j][operation[j]]

            start = max(job_time[j], machine_time[machine])
            finish = start + duration

            heuristic = 1 / (finish + 0.001)

            value = (
                pheromone[previous][j] ** alpha *
                heuristic ** beta
            )

            probability.append(value)

        probability = np.array(probability)
        probability /= probability.sum()

        selected = random.choices(
            feasible,
            weights=probability,
            k=1
        )[0]
```

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        machine, duration = jobs[selected][operation[selected]]

        start = max(job_time[selected], machine_time[machine])
        finish = start + duration

        schedule.append({
            'Job': selected + 1,
            'Operation': operation[selected] + 1,
            'Machine': machine + 1,
            'Start': start,
            'Finish': finish,
            'Duration': duration
        })

        job_time[selected] = finish
        machine_time[machine] = finish

        operation[selected] += 1

        sequence.append(selected)
        previous = selected

    makespan = max(job_time)

    return schedule, makespan, sequence


def update_pheromone(sequence, makespan):

    global pheromone

    pheromone *= (1 - rho)

    previous = num_jobs

    for job in sequence:
        pheromone[previous][job] += Q / makespan
        previous = job


for iteration in range(iterations):

    iteration_best = float('inf')
    best_sequence = None

    for ant in range(num_ants):

        schedule, makespan, sequence = generate_schedule()

        if makespan < best_makespan:
            best_makespan = makespan
            best_schedule = schedule

        if makespan < iteration_best:
            iteration_best = makespan
            best_sequence = sequence

    update_pheromone(best_sequence, iteration_best)


print("\nOPTIMIZED JOB SCHEDULE")

df = pd.DataFrame(best_schedule)
df = df.sort_values(by=['Start', 'Machine'])

print(df.to_string(index=False))

print("\nMinimum Makespan:", best_makespan)

print("\nMACHINE UTILIZATION")

total_processing_time = sum(
    duration
    for job in jobs.values()
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```
    for machine, duration in job
)

for m in range(num_machines):

    busy_time = sum(
        item['Duration']
        for item in best_schedule
        if item['Machine'] == m + 1
    )

    utilization = busy_time / best_makespan * 100

    print(
        f"Machine {m + 1}: "
        f"Busy Time = {busy_time}, "
        f"Utilization = {utilization:.2f}%"
    )

overall_utilization = (
    total_processing_time /
    (num_machines * best_makespan)
) * 100

print("\nOverall Machine Utilization:",
      round(overall_utilization, 2), "%")
```

```
OPTIMIZED JOB SCHEDULE
 Job  Operation  Machine  Start  Finish  Duration
   1          1        1      0       3         3
   2          1        2      0       2         2
   3          1        3      0       2         2
   2          2        3      2       5         3
   3          2        1      3       7         4
   1          2        2      3       5         2
   1          3        3      5       9         4
   2          3        1      7       9         2
   3          3        2      7      10         3

Minimum Makespan: 10

MACHINE UTILIZATION
Machine 1: Busy Time = 9, Utilization = 90.00%
Machine 2: Busy Time = 7, Utilization = 70.00%
Machine 3: Busy Time = 9, Utilization = 90.00%

Overall Machine Utilization: 83.33 %
```